UniCIC: A Unified Collaborative Inference Framework for Remote Sensing Change Detection and Captioning

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***Abstract*—Multilevel change interpretation (MCI) aims to jointly model change detection (CD) and change captioning (CC) within a unified framework, integrating pixel-level localization and semantic-level description to achieve multigranularity change understanding. However, existing MCI methods suffer from two critical drawbacks: 1) they lack a mechanism to separately capture fine-grained spatial cues for CD and object-level semantic cues for CC during feature encoding and 2) they lack effective cross-task interaction mechanisms, which severely restricts the collaborative performance gain. To address these issues, we propose unified collaborative inference framework (UniCIC) that constructs heterogeneous change representations from bitemporal images for different tasks and enables cross-task collaborative inference. Specifically, we propose an SAM-prior-guided task-oriented perception encoding (SP-TOPE) module, which employs an SAM-based region-level perception frame to capture the object-level semantics required for CC and introduces a randomly initialized perception frame to aggregate the fine-grained spatial change cues required for CD, leveraging the cross-frame modeling capability of a video encoder to generate task-oriented change representations. Furthermore, we design a spatial–semantic mutual guidance (S^2MG) mechanism, which establishes a sequential cross-task interaction pathway by first using a coarse change mask to provide spatial priors for caption generation and then feeding semantic entities extracted from the generated description back into the detection branch to refine the final change mask, thereby improving the consistency between change localization and semantic description. Comprehensive evaluations on the LEVIR-MCI dataset show that the proposed method achieves competitive results in both CD and CC metrics. The code is avaliable at https://github.com/Jiahuiqu/UniCIC.**

***Index Terms*—Change interpretation, multitask learning (MTL), mutual guidance, video encoding (VE).**

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I. Introduction

REMOTE sensing images [1], [2], [3], with their unique macroscopic observation perspective, have become a core technical means for monitoring and analyzing the spatiotemporal dynamic evolution of the Earth's surface [4], [5], [6]. As a major research direction in remote sensing, remote sensing image change interpretation (RSICI) [7], [8] identifies and analyzes variations in multitemporal remote sensing images covering the same geographic region and can provide dependable data support and technical safeguards for scientific decision. It has been widely applied in critical applications including environmental protection, resource management, and other fields [9].

Existing RSICI tasks mainly consist of two core components: change detection (CD) and change captioning (CC). As shown in Fig. 1(a) and (b), CD focuses on identifying the spatial locations and extents of surface changes at the pixel level. Unlike traditional visual interpretation tasks, such as object detection [10], [11] and image classification [12], [13], which primarily analyze information within the image modality, CC [14] further employs natural language to describe the semantic attributes and practical implications of these changes. Although both techniques have achieved considerable progress, relying on a single task paradigm is insufficient to achieve comprehensive understanding of changes. CD provides accurate spatial localization but fails to deeply mine the semantic information behind changes. In contrast, although CC provides rich semantic interpretations, it generally cannot localize changes accurately at the pixel level. Therefore, it is urgent to develop a multilevel change interpretation (MCI) method that enables fine-grained change localization while performing in-depth semantic analysis, so as to provide more comprehensive and reliable support for remote sensing image change analysis.

Multitask learning (MTL) jointly optimizes several mutually related tasks under a unified framework. By sharing feature knowledge through correlations among different tasks to mutually boost model performance, it has, thus, become a mainstream and feasible research direction for integrating the two aforementioned tasks. The benchmark presented in [15] incorporates the LEVIR-MCI dataset and the multistage MCINet model, enabling CC and CD to be jointly implemented within one network architecture and providing

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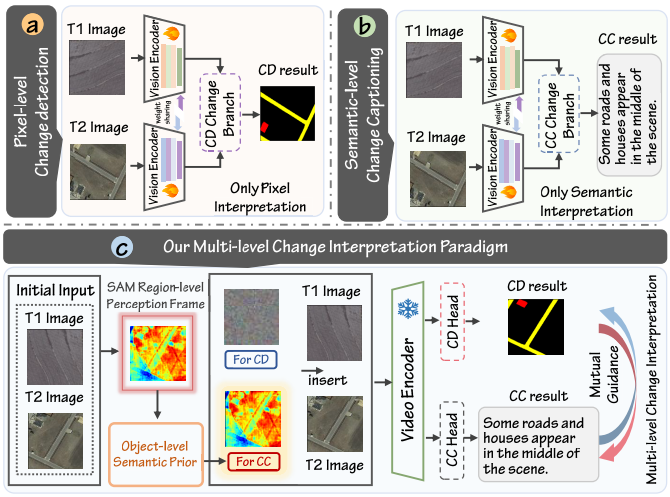


Fig. 1. Architecture comparison of our method with the CD task and the CC task. (a) Pixel-level CD. (b) Semantic-level CC. (c) Our MCI paradigm.

a valuable reference for subsequent research. On this basis, Wang et al. [16] further proposed a unified framework that models the extracted bitemporal deep features to generate generic change-aware representations suitable for two tasks. Although the aforementioned methods have achieved certain performance improvements, they still suffer from critical limitations in feature modeling and task collaboration dimensions, which hinder further breakthroughs in multitask change interpretation performance.

1) Existing MCI methods typically introduce task differentiation only after bitemporal feature encoding, hindering the direct learning of task-oriented change representations tailored to spatial localization for CD and changed-object semantic interpretation for CC during encoding.
2) Existing MTL methods mostly adopt a decoupled-branch paradigm, where CD and CC are modeled independently. These methods mainly rely on loss functions to realize implicit constraints. Although such methods can output two types of predictions simultaneously, these outputs are generally treated as mutually independent task results without explicit interaction mechanisms during inference.

Therefore, how to accommodate the heterogeneous information requirements of CD and CC within a unified encoding process, while establishing effective cross-task interactions for collaborative optimization, remains a critical challenge. Zhu et al. [17] introduced Change3D, a video encoding (VE) paradigm based on perception frames, and applied it separately to the individual CC and CD tasks. Inspired by this idea, we focus on the differentiated representation requirements of the two tasks in the unified encoding process: CD relies on fine-grained spatial localization cues, while CC focuses more on region-level semantic information. Due to the prominent divergence in their information preferences, a single homogeneous perception frame struggles to generate change representations tailored to each task. Accordingly, constructing heterogeneous change information carriers for spatial localization and semantic interpretation is essential for the unified encoding of CD and CC. In addition, CD and CC exhibit an inherent complementarity. The spatial localization results produced by the former can provide regional constraints for the latter, thereby reducing interference from irrelevant backgrounds; meanwhile, the semantic entities identified by the latter can, in turn, enhance the former's ability to distinguish change objects. Therefore, establishing a cross-task interaction pathway can help fully exploit the collaborative benefits of MTL.

Based on the above analysis, we develop unified collaborative inference framework (UniCIC), a unified framework whose architecture is illustrated in Fig. 1(c). The framework consists of an SAM-prior-guided task-oriented perception encoding (SP-TOPE) module and a spatial–semantic mutual guidance (S^2MG) module. To address the distinct information requirements of CD and CC, SP-TOPE incorporates an SAM-based region-level perception frame for CC and a randomly initialized perception frame for CD into the bitemporal image sequence, enabling video modeling to form task-oriented change representations from object-level semantic priors and fine-grained spatial cues. Subsequently, S^2MG establishes connections between detection results and caption semantics, allowing spatial localization information to contribute to caption generation and semantic information to assist in refining change regions. Through these designs, UniCIC jointly accommodates pixel-level change localization and semantic-level change interpretation within a unified framework while promoting the joint optimization of both tasks through cross-task information collaboration. The main contributions of our work are summarized as follows.

1) We propose a UniCIC that integrates task-specific change representation learning and cross-task collaborative optimization into a unified architecture for jointly modeling CD and CC.
2) We design an SP-TOPE module, which leverages SAM-based object-level priors to construct a region-aware perception frame for CC and introduces a randomly initialized frame for CD, enabling task-oriented change representation learning during unified encoding.
3) We propose an S^2MG module, which reformulates CD and CC from independent prediction tasks into a mutually driven joint reasoning process, improving multitask collaboration performance.

The remainder of this article is structured as follows. Section II reviews related studies on remote sensing change interpretation, MTL, and video encoders. Section III presents the proposed framework and its main components. Section IV describes the dataset and experimental settings and analyzes the results. Finally, Section V concludes this article and discusses future research directions.

II. Related Work

A. Remote Sensing Image Change Interpretation

RSICI mainly consists of two core research tasks: CD and CC. Early CD methods were primarily developed based on classical techniques such as change vector analysis, principal

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component analysis, and postclassification comparison. In recent years, advances in deep learning have substantially accelerated methodological innovation in this field. Convolutional neural networks, Transformers, and Mamba-based models constitute the mainstream architectures widely adopted in this field. Regarding CNN-based approaches, Daudt et al. [18] developed three architectures based on fully convolutional networks (FCNs). For multiscale feature extraction, Peng et al. [19] designed a multiside output fusion strategy based on an improved UNet++ to enhance multiscale feature extraction capability. Li et al. [20] developed a temporal feature interaction module to improve the information flow efficiency among multiscale bitemporal features. For the Mamba-based methods, Qu et al. [21] proposed MambaFedCD, which combines spatial–spectral–temporal modeling, diversity-aware aggregation, and active learning for privacy-preserving hyperspectral CD with limited labels. For the Transformer-based methods, Ding et al. [22] constructed an encoder framework, where self-attention captures long-range dependencies between bitemporal features to obtain a global spatiotemporal receptive field. Chen et al. [23] employed a hybrid head Transformer to jointly model cross-correlations and self-correlations of bitemporal features, enhancing the discriminability of change representations.

While traditional visual interpretation tasks [24], [25], [26] primarily focus on visual feature discrimination, CC aims to establish connections between bitemporal visual changes and natural language to generate semantic descriptions of land-cover changes. With advances in deep learning, the encoder–decoder architecture has become the mainstream paradigm for this task. Jhamtani and Berg-Kirkpatrick [27] first formulated this task and adopted a Siamese CNN-RNN architecture to represent the differences between similar images. To effectively distinguish interference from real change regions, Park et al. [28] proposed the dual dynamic attention (DUDA) model, thereby substantially enhancing the accuracy and robustness of generated captions. Liu et al. [29] established the large-scale remote sensing image dataset LEVIR-CC and proposed a dual-branch Transformer encoder that leverages bitemporal feature differences to enhance the image feature representation of each temporal phase. Chang and Ghamisi [30] proposed an attentive changes-to-captions (Chg2Cap) network, which locates key change-related features via hierarchical self-attention modules. To address the insufficient joint spatial–temporal modeling, Liu et al. [31] introduced RSCaMa, in which multilayer CaMa modules are employed to efficiently model spatial and temporal information jointly.

Although the aforementioned single-task methods have achieved remarkable progress in both CD and CC, relying solely on a single-task paradigm is insufficient to realize comprehensive interpretation of remote sensing image changes. Accordingly, developing MCI methods that accomplish in-depth semantic analysis while achieving fine-grained change localization, so as to provide more comprehensive and complete interpretation support for remote sensing change analysis, has become an urgent demand in this field.

B. Multitask Learning

MTL is an important joint learning paradigm in the field of deep learning. It leverages the intrinsic relationships among tasks to achieve feature sharing and information complementarity, thereby improving the learning performance of individual tasks. Research on MTL has permeated multiple technical directions, among which mutual guidance among tasks is one of the key technologies for enhancing MTL performance, and related research has achieved significant results in various fields. Tian et al. [32] proposed the cyclic colearning (CCoL) paradigm, establishing a colearning cycle for sounding object visual grounding and audio–visual sound separation, making them mutually beneficial. Xing and Tsang [33] developed coguiding Net, whose bidirectional interaction mechanism enables intent detection and slot filling to guide each other and undergo collaborative optimization. He et al. [34] proposed the cross-task policy guidance (CTPG) framework, providing explicit guidance for unmastered tasks through the control policies of already proficient tasks, thereby achieving knowledge transfer among multiple tasks.

In the field of remote sensing image processing, MTL has gradually been applied to change interpretation tasks. Current mainstream research mainly focuses on jointly training CD and CC tasks, achieving simultaneous predictions for both tasks by sharing the underlying feature extractor. Liu et al. [15] proposed the MCI model, following a shared-separation architecture, wherein only the image encoder was shared at the lower level to extract feature representations, while task-specific branches were employed at the upper level to achieve independent predictions for each task. Yang et al. [35] developed a key-change-feature and instruction-tuned framework, where a key feature perceiver extracts shared change representations from bitemporal images. Although MTL has made progress in the field of remote sensing change interpretation, existing models mostly only achieve feature sharing and joint training of the two tasks, fail to effectively integrate the mutual guidance mechanism, and cannot realize two-way optimization through the complementary information between tasks, resulting in the insufficient exertion of the synergistic effect.

C. Video Encoder

Video understanding requires jointly modeling spatial appearance and temporal variations in sequential visual data to support tasks such as action recognition and action detection. Among existing approaches, 3-D convolutional neural networks (3-D CNNs) extend conventional spatial convolutional kernels along the temporal dimension, enabling the simultaneous encoding of local spatial structures and interframe dynamics. Although early 3-D CNNs typically incurred high computational costs, subsequent architectures achieved considerable improvements in trainability and computational efficiency. For example, I3D initializes 3-D convolutional kernels by inflating pretrained 2-D kernels [36], whereas X3D progressively expands selected network dimensions to balance recognition accuracy and computational complexity [37]. These studies demonstrate the effectiveness of 3-D CNNs

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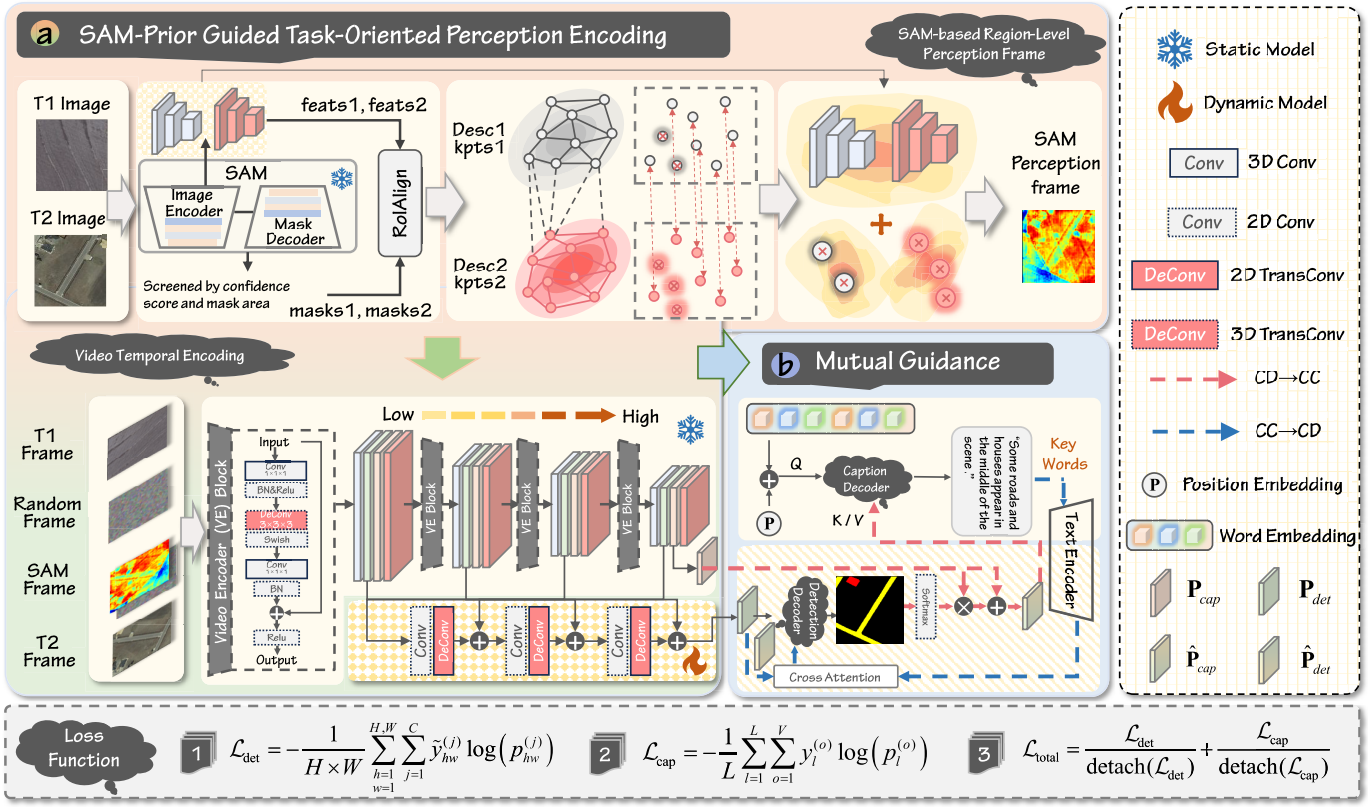


Fig. 2. Overall architecture of UniCIC that comprises the SP-TOPE module and the S^{2}MG module. SP-TOPE generates task-oriented change representations by jointly encoding the bitemporal images, an SAM-based perception frame for CC, and a randomly initialized perception frame for CD. S^{2}MG establishes sequential cross-task interaction by using the coarse change mask to guide caption generation and then feeding semantic entities from the generated caption back into the CD branch. (a) SAM-prior guided task-oriented perception encoding module. (b) Spatial-semantic mutual guidance module.

in jointly encoding spatial context and short-range temporal variations. Accordingly, we employ X3D as the video encoder to extract basic features from bitemporal images because of its established spatiotemporal encoding capability.

III. Methodology

A. Problem Formulation

Given bitemporal remote sensing images $\mathbf{I}_1, \mathbf{I}_2 \in \mathbb{R}^{H\times W\times C}$, where H and W stand for spatial dimensions and C is the number of spectral channels, the unified MCI framework outputs the equal-resolution change probability map $\mathbf{M}$ and change description sentences S. Current MCI methods follow a pipeline of a shared image encoder, task-specific difference extraction, and task-dedicated decoding, which is formally defined as

$$\begin{cases} \mathcal{R}_c\left(\mathcal{T}_c\left(\Phi_{\mathrm{I}}\left(\mathbf{I}_1;\mathbf{I}_2\right)\right)\right) \longrightarrow \mathrm{S} \\ \mathcal{R}_d\left(\mathcal{T}_d\left(\Phi_{\mathrm{I}}\left(\mathbf{I}_1;\mathbf{I}_2\right)\right)\right) \longrightarrow \mathbf{M} \end{cases} \tag{1}$$

where $\Phi_{\mathrm{I}}(\cdot)$ denotes the shared image encoder and $\mathcal{T}_d(\cdot)$ and $\mathcal{T}_c(\cdot)$ represent the difference feature extraction branches for two tasks, while $\mathcal{R}_d(\cdot)$ and $\mathcal{R}_c(\cdot)$ correspond to the CD head and the CC head, respectively.

As can be seen from (1), existing MCI models first perform independent feature extraction via the shared image encoder $\Phi_{\mathrm{I}}(\cdot)$. As low-level image encoding is a task-agnostic general feature modeling process, it cannot explicitly characterize the feature differences of changed regions. Hence, two dedicated differential feature extraction branches $\mathcal{T}_d(\cdot)$ and $\mathcal{T}_c(\cdot)$ must be additionally introduced to meet the requirements of downstream detection and captioning tasks. Furthermore, the change probability map $\mathbf{M}$ and change caption sequence S output by their respective decoding heads are mutually isolated. The complementary characteristics of the two prediction outputs have not been fully exploited, which limits the performance of multitask collaborative inference. Accordingly, we develop a unified collaborative inference framework based on heterogeneous representations. It introduces heterogeneous perception frames into unified VE to implicitly capture cross-temporal differences and learn task-specific representations for spatial localization and semantic interpretation. The model further performs sequential bidirectional spatial-semantic guidance, where change regions guide caption generation and caption semantics are fed back to refine detection, promoting complementary learning and joint optimization of both tasks. The overall process is formulated as follows:

$$\begin{cases} \mathcal{R}_c\left(\Phi\left(\mathbf{I}_1;\mathbf{I}_2\right)\right) \xrightarrow{\mathcal{H}(\cdot),\mathcal{R}_d(\Phi(\mathbf{I}_1;\mathbf{I}_2))} \mathrm{S} \\ \mathcal{R}_d\left(\Phi\left(\mathbf{I}_1;\mathbf{I}_2\right)\right) \xrightarrow{\mathcal{H}(\cdot),\mathrm{S}} \mathbf{M} \end{cases} \tag{2}$$

where $\Phi(\cdot)$ denotes the SP-TOPE module, which implicitly models cross-temporal differential features during the encoding stage, and $\mathcal{H}(\cdot)$ refers to the S^{2}MG mechanism that constructs cross-task interaction pathways at the prediction level.

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B. Overall Architecture

As illustrated in Fig. 2, UniCIC takes bitemporal remote sensing images $\mathbf{I}_1$ and $\mathbf{I}_2$ as input and mainly consists of an SP-TOPE module and an S^{2}MG module. Specifically, SP-TOPE leverages SAM-based object-level priors to construct a region-aware perception frame for CC and introduces a randomly initialized learnable perception frame for CD. Together with the bitemporal images, the two frames form a temporal input sequence, where cross-frame interactions aggregate bitemporal change information into the respective perception frames, yielding task-oriented change representations. On this basis, S^{2}MG performs sequential spatial-semantic interaction at the prediction stage. The CD branch first generates a coarse change mask to provide a spatial prior for change caption generation. Subsequently, semantic entity features extracted from the generated caption are fed back into the CD branch to further refine the change mask. Finally, the model outputs the refined CD result and its corresponding change caption.

C. SAM-Prior-Guided Task-Oriented Perception Encoding

Existing MCI methods typically perform task-specific change modeling only after separately encoding bitemporal images, leaving task differentiation largely separate from feature encoding. To address this limitation, we propose the SP-TOPE module. The two perception frames adopt different initialization strategies according to the distinct requirements of CD and CC. CC relies on object-level semantics to describe changes; therefore, we construct its perception frame using SAM-derived regional features, which provide candidate changed region and contextual priors. In contrast, CD requires fine-grained pixel-level localization. We, thus, use a randomly initialized learnable frame without a predefined regional partition, allowing it to learn spatial change cues through joint VE and CD supervision. Together with the bitemporal images, these heterogeneous perception frames form a temporal input sequence. Through cross-frame interactions in the video encoder, each frame selectively aggregates task-relevant change information and implicitly models cross-temporal differences, yielding task-oriented representations for semantic interpretation and spatial localization within unified encoding. The overall procedure is formulated as follows:

$$\Phi(\mathbf{I}_1;\mathbf{I}_2) = \mathcal{F}_{VTE}\left\{\mathbf{I}_1;\mathbf{I}^P_{Random};\mathcal{F}_{SAM_{RL}}(\mathbf{I}_1,\mathbf{I}_2);\mathbf{I}_2\right\} \tag{3}$$

$$\mathbf{I}^P_{SAM} = \mathcal{F}_{SAM_{RL}}(\mathbf{I}_1,\mathbf{I}_2) \tag{4}$$

$$\{\mathbf{P}_{det}\};\mathbf{P}_{cap} = \mathcal{F}_{VTE}\left\{\mathbf{I}_1;\mathbf{I}^P_{Random};\mathbf{I}^P_{SAM};\mathbf{I}_2\right\} \tag{5}$$

where $\mathcal{F}_{SAM_{RL}}(\cdot)$ denotes the SAM-based region-level perception frame submodule, $\mathcal{F}_{VTE}(\cdot)$ denotes the video temporal encoding submodule, and $\mathbf{I}^P_{SAM} \in \mathbb{R}^{H\times W\times 3}$ and $\mathbf{I}^P_{Random} \in \mathbb{R}^{H\times W\times 3}$ represent the SAM-based region-level perception frame and the randomly initialized perception frame, respectively. $\{\mathbf{P}_{det}\}$ and $\mathbf{P}_{cap}$ denote the preliminary difference representations generated by the SP-TOPE module.

1) SAM-Based Region-Level Perception Frame: As illustrated in Fig. 2(a), for each temporal image $\mathbf{I}_t \in \mathbb{R}^{H\times W\times 3}$ ($t \in (1,2)$), we feed it into the pretrained SAM model to obtain the global feature maps $\mathbf{S}_t$ and the segmentation masks $\mathcal{M}_t$. To guarantee mask quality, we perform preliminary filtering on the generated masks, thereby obtaining two sets of high-quality regional masks $\mathcal{M}'_t$ and their corresponding bounding boxes $\mathcal{B}_t$

$$\mathbf{S}_t = \mathcal{F}^{\text{SAM}}_{\text{enc}}(\mathbf{I}_t), \mathcal{M}_t = \mathcal{F}^{\text{SAM}}_{\text{dec}}(\mathbf{S}_t), t \in \{1,2\} \tag{6}$$

$$\mathcal{M}'_t = \{m \in \mathcal{M}_t | \text{score}(m) > \tau_s \wedge \text{area}(m) > \tau_a\} \tag{7}$$

$$\mathcal{B}_t = \left\{\mathcal{B}box(m) \,\middle|\, m \in \mathcal{M}'_t\right\} \tag{8}$$

where $\mathcal{F}^{\text{SAM}}_{\text{enc}}$ denotes the SAM image encoder, $\mathcal{F}^{\text{SAM}}_{\text{dec}}$ denotes the SAM mask decoder, τ_s and τ_a denote the confidence threshold and the area threshold, respectively, and $\mathcal{B}box(\cdot)$ denotes the mapping function that generates a bounding box from a mask.

Based on $\mathbf{S}_t$, we employ RoIAlign to align the local features corresponding to each regional bounding box and map them into fixed-dimensional regional feature descriptors $d_{t,b}$ while recording the regional center keypoints $k_{t,b}$ of each region. With the keypoints as position constraints and the descriptors as feature inputs, we compute a matching matrix $H \in \mathbb{R}^{|\mathcal{B}_1|\times|\mathcal{B}_2|}$ and filter out irrelevant regions, and the finally obtained differential regions are the candidate changed ground objects in the scene

$$d_{t,b} = \mathcal{F}_{\text{RoI}}(\mathbf{S}_t, b), \quad k_{t,b} = \mathcal{F}_{\text{center}}(b) \quad \forall\, b \in \mathcal{B}_t \tag{9}$$

$$D_t = \left\{d_{t,b} \,\middle|\, b \in \mathcal{B}_t\right\}, \quad K_t = \left\{k_{t,b} \,\middle|\, b \in \mathcal{B}_t\right\} \tag{10}$$

$$H = \mathcal{F}(D_1, K_1, D_2, K_2) \tag{11}$$

$$\mathcal{U}_1 = \left\{b_x \in \mathcal{B}_1 \,\middle|\, \max_y H_{x,y} < \tau_h\right\} \tag{12}$$

$$\mathcal{U}_2 = \left\{b_y \in \mathcal{B}_2 \,\middle|\, \max_x H_{x,y} < \tau_h\right\} \tag{13}$$

where $H_{x,y}$ denotes the matching confidence between the xth region at $t = 1$ and the yth region at $t = 2$, τ_h represents the matching score threshold, and $\mathcal{U}_1$ and $\mathcal{U}_2$ represent the sets of unmatched regions at time $t = 1$ and $t = 2$.

Finally, according to the mask locations, the retained regional features without matches are mapped back to a feature map whose size is identical to that of $\mathbf{I}_t$, and fused with $\mathbf{S}_t$ to supplement contextual information, thus generating the SAM-based region-level perceptual frame $\mathbf{I}^P_{SAM}$. This frame encodes change priors and ground object semantics and serves as a feature register to guide the subsequent temporal encoding to efficiently learn differential features

$$\mathbf{F}^P_{SAM} = \alpha \frac{1}{N_p} \sum_{b\in\mathcal{U}_1\cup\mathcal{U}_2} d_{t,b} + (1-\alpha)\,\mathcal{F}_{\text{proj}}\left(\frac{\mathbf{S}_1+\mathbf{S}_2}{2}\right) \tag{14}$$

$$\mathbf{I}^P_{SAM} = Conv_{1\times 1}\left(\mathbf{F}^P_{SAM}\right) \tag{15}$$

where N_p denotes the number of regions covering each spatial location, which is used for feature averaging to ensure consistency, and α is a learnable fusion weight that balances regional features and contextual information.

2) Video Temporal Encoding: To address the limitation of region-level priors in modeling fine-grained pixel-level changes, we initialize a random perception frame $\mathbf{I}^P_{Random}$ for the CD task. Given the constructed multiframe sequence $\mathcal{I} = \{\mathbf{I}_1, \mathbf{I}^P_{Random}, \mathbf{I}^P_{SAM}, \mathbf{I}_2\}$, we employ a hierarchical video encoder [37] to model the temporal features. In Fig. 2(a), the multiframe sequence is concatenated along the temporal

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dimension and then passed through multiple VE blocks to produce multiscale hierarchical features $\{\mathbf{F}_i^t\}$. Here, the sequence length is t = 4, and $\mathbf{F}_i^t$ denotes the feature generated by the ith encoder layer for the tth frame.

To accommodate the distinct requirements of the two tasks, we adopt different feature selection strategies. Since CD requires pixel-level spatial localization, shallow features provide fine-grained spatial details for delineating small changed regions, while deep features capture contextual information for recognizing large-scale structural changes. Therefore, we extract the second-frame features $\{\mathbf{F}_0^2, \mathbf{F}_1^2, \mathbf{F}_2^2, \mathbf{F}_3^2\}$ from all encoder layers as the CD perception features $\{\mathbf{P}_{det}\}$. In contrast, CC primarily relies on object-level semantics and scene context to generate change descriptions. Therefore, the third-frame feature $\mathbf{F}_3^3$ from the top encoder layer is selected as the CC perception feature $\mathbf{P}_{cap}$, which incorporates regional priors and high-level contextual information. The process can be expressed as

$$\left\{\mathbf{F}_i^1; \mathbf{F}_i^2; \mathbf{F}_i^3; \mathbf{F}_i^4\right\} = \mathcal{F}_{VTE}\left\{\mathbf{I}_1; \mathbf{I}_{Random}^P; \mathbf{I}_{SAM}^P; \mathbf{I}_2\right\} \tag{16}$$

$$\{\mathbf{P}_{det}\} = \left\{\mathbf{F}_i^2\right\}, \quad \mathbf{P}_{cap} = \mathbf{F}_3^3. \tag{17}$$

D. Spatial-Semantic Mutual Guidance Module

Existing approaches generally process CD and CC as two independent tasks. The absence of effective cross-task interaction prevents spatial localization and semantic understanding from complementing each other, which can result in inconsistencies between caption text and change masks. To this end, we design an S^{2}MG module, which achieves collaborative optimization between tasks by constructing a bidirectional information transmission mechanism. Considering the differences in output generation, we select CD as the initial source of cross-task guidance. CD produces a change probability map through parallel decoding, whereas CC generates a complete description autoregressively before semantic entities are extracted for feedback. This motivates a sequential interaction strategy in which CD first guides CC, followed by semantic feedback to refine CD. Specifically, the CD representation $\{\mathbf{P}_{det}\}$ produced by the SP-TOPE module is first decoded to obtain a coarse-grained change probability map, which serves as a spatial prior to enhance the captioning representation $\mathbf{P}_{cap}$. This spatial guidance helps the captioning decoder focus on potential change regions and generate more accurate change descriptions. Subsequently, key category-level semantic cues are extracted from the generated captions and fed back into the CD branch to improve semantic discrimination and refine the localization of changed regions, yielding the final change mask

$$\hat{\mathbf{P}}_{cap} = \mathcal{H}\left(\mathcal{R}_d\left\{\mathbf{P}_{det}\right\}, \mathbf{P}_{cap}\right) \tag{18}$$

$$\hat{\mathbf{P}}_{det} = \mathcal{H}\left(\mathcal{R}_c\left(\hat{\mathbf{P}}_{cap}\right), \mathbf{P}_{det}\right) \tag{19}$$

$$\mathrm{S} = \mathcal{R}_c\left(\hat{\mathbf{P}}_{cap}\right), \quad \mathbf{M} = \mathcal{R}_d\left(\hat{\mathbf{P}}_{det}\right) \tag{20}$$

where $\hat{\mathbf{P}}_{cap}$ and $\hat{\mathbf{P}}_{det}$ denote the final difference representations processed by the S^{2}MG module and $\hat{\mathbf{P}}_{cap}$ is fed into the CC decoder to yield the change description result S, while $\hat{\mathbf{P}}_{det}$ passes through the CD decoder to generate the change mask $\mathbf{M}$.

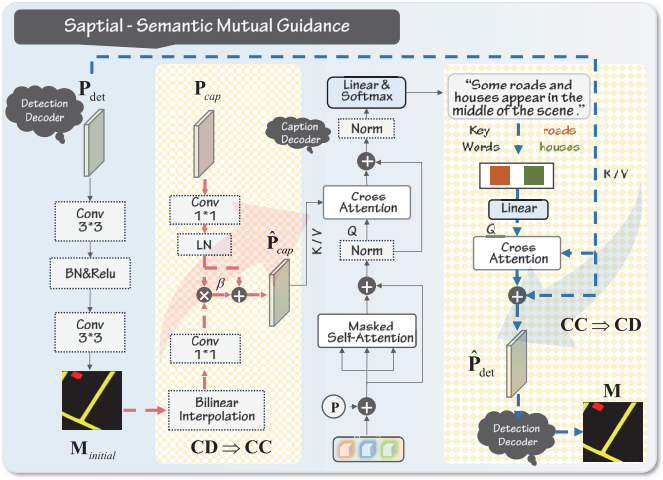


Fig. 3. Architecture of the S^{2}MG module.

1) Change Detection and Captioning Decoders: As shown in Fig. 3, the CD decoder receives $\{\mathbf{F}_i^t\}, (t = 2)$, generated by SP-TOPE as its input. It progressively integrates change information across different scales and restores the spatial resolution of the features in a deep-to-shallow manner. To distinguish the task-specific features produced during encoding from the aggregated features obtained during decoding, $\mathbf{F}_i^t$ and $\mathbf{D}_{\mathrm{I}-i}^t$ are, respectively, used to denote the CD perception frame feature and the decoded feature at the ith level. $i \in \{1, \ldots, \mathrm{I}\}$, and I denotes the deepest layer of the encoder. At the ith decoding stage, the decoded feature from the preceding stage is successively projected in the channel dimension and upsampled, and is then fused with the CD perception frame feature at the current scale

$$\mathbf{D}_{\mathrm{I}-i}^t = \mathbf{F}_{i-1}^t + \mathcal{US}_i\left(\mathcal{C}_i\left(\mathbf{F}_i^t\right)\right), \quad i = \mathrm{I}, \ldots, 1 \tag{21}$$

where $\mathcal{C}_i(\cdot)$ denotes the channel mapping operation at the ith decoding stage and $\mathcal{US}_i(\cdot)$ represents the corresponding up-sampling operation. Consequently, the resulting shallowest decoded feature $\mathbf{P}_{det} = \mathbf{D}_2^2$ integrates multiscale change semantics with spatial localization information. It is subsequently fed into a 3×3 convolutional layer for spatial refinement, after which an argmax operation is applied along the channel dimension to assign a change category to each pixel and generate the CD map $\mathbf{M}$ with C categories

$$\mathbf{M} = \arg\max\left(\mathrm{Conv}_{3\times3}\left(\mathbf{P}_{det}\right)\right), \quad \mathbf{M} \in \{0, \ldots, C-1\}^{H\times W}. \tag{22}$$

A conventional Transformer architecture is adopted for the CC decoder. In the training stage, masked multihead self-attention restricts each token to attending only to earlier tokens, thereby preserving the autoregressive nature of caption generation. Subsequently, encoder–decoder attention (EDAtt) enables the decoder to selectively retrieve information from the fused change representation according to the current linguistic context. Specifically, the output of the preceding masked self-attention module is projected to form the query (Q), whereas the keys (K) and values (V) are generated through separate

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linear projections of the CC features. The resulting cross-modal interaction is formulated as follows:

$$EDAtt\left(\mathbf{S}^{l}, \mathbf{P}_{cap}\right) = Concat\left(head_1, \ldots, head_h\right)\mathbf{W}_o \tag{23}$$

$$head_m = CrossAtt\left(\mathbf{S}^{l}\mathbf{W}_m^{Q}, \mathbf{P}_{cap}\mathbf{W}_m^{K}, \mathbf{P}_{cap}\mathbf{W}_m^{V}\right) \tag{24}$$

where $\mathbf{W}_m^{Q}, \mathbf{W}_m^{K}, \mathbf{W}_m^{V} \in \mathbb{R}^{d\times d}$ are trainable matrices of the mth head, $\mathbf{W}_o \in \mathbb{R}^{hd\times d}$ is a trainable matrix to perform linear projection changing the feature channel dimension, $\mathbf{S}^{l} \in \mathbb{R}^{L\times d}$ is the sentence embedding from the masked self-attention sublayer, L is the length of the sentence, and h is the head number of the MCA.

2) Change Detection and Captioning Mutual Guidance: As shown in Fig. 3, the CD feature $\{\mathbf{P}_{det}\}$ is first processed by the CD decoder $\mathcal{R}_d(\cdot)$ to generate a coarse-grained probability map $\mathbf{M}_{initial} \in \mathbb{R}^{H\times W\times 3}$. Its three channels correspond to the pixel-level probabilities of road change, building change, and background, respectively. To align the spatial resolution with that of the CC feature $\mathbf{P}_{cap}$, we first downsample the probability map channelwise using bilinear interpolation, thereby preserving the spatial distribution and category information of the change regions. Subsequently, a 1×1 convolution projects the three probability channels to the channel dimension of $\mathbf{P}_{cap}$, followed by a Sigmoid activation that constrains the weight values to (0, 1). Consequently, the resulting weight map has the same spatial resolution and channel dimension as $\mathbf{P}_{cap}$, enabling elementwise feature modulation. The above procedure is mathematically expressed as follows:

$$\mathbf{M}_{initial} = \mathcal{R}_d\left\{\mathbf{P}_{det}\right\} \tag{25}$$

$$\mathbf{W} = \sigma\left(\mathrm{Conv}_{1\times 1}\left(\mathcal{F}_{\downarrow}\left(\mathbf{M}_{initial}\right)\right)\right) \tag{26}$$

where $\mathcal{F}_{\downarrow}(\cdot)$ represents bilinear interpolation and $\sigma(\cdot)$ denotes the sigmoid activation function. Finally, elementwise weighted modulation $\mathbf{W}$ is applied to $\mathbf{P}_{cap}$. This operation enhances feature responses in change regions and suppresses background interference at the spatial level while injecting category priors of roads and buildings to guide the caption generation process to precisely focus on change regions and corresponding land cover types. On this basis, we introduce a learnable fusion weight β combined with a residual connection, so as to deeply integrate the spatial localization information extracted by the CD branch into the feature encoding process of the CC task. The final spatial prior-enhanced CC feature is obtained by

$$\hat{\mathbf{P}}_{cap} = LN\left(\beta\cdot\mathbf{P}_{cap}\odot\mathbf{W} + \mathbf{P}_{cap}\right). \tag{27}$$

In addition to the CD-guided CC branch, we further design a CD feature enhancement branch based on semantic guidance from core change land cover keywords, which leverages the high-level semantic information output by the CC task to reversely optimize the pixel-level change localization accuracy of the CD branch. First, we feed the aforementioned spatial prior-enhanced CC feature $\hat{\mathbf{P}}_{cap}$ into the CC decoder $\mathcal{R}_c(\cdot)$ to generate the corresponding change caption sentence S. In change caption sentences, the noun components in subject and object positions correspond to the core land-cover entities undergoing change, which are the most discriminative semantic units for change regions. Based on this observation, we extract core change entities from the generated change caption sentence S to form an entity set $\mathcal{E} = \{w_1, w_2, \ldots, w_i, \ldots, w_N\}$ and map each entity w_i into the word embedding space to encode it as a discrete word vector representation. To generate a unified semantic query vector, we apply average pooling over the extracted multiple change entity word embeddings

$$\mathcal{R}_c\left(\hat{\mathbf{P}}_{cap}\right) \rightarrow \mathcal{E} = \{w_1, w_2, \ldots, w_i, \ldots, w_N\}$$

$$\bar{e} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{E}\left[w_i\right] \tag{28}$$

where $\bar{e}$ represents the average embedding aggregated from the word embeddings of core change entities extracted from change captions, which is further projected to generate the semantic query e_{text}

$$e_{text} = \boldsymbol{W}_p\cdot\bar{e} + \boldsymbol{b}_p \tag{29}$$

where $\boldsymbol{W}_p$ and $\boldsymbol{b}_p$ are the learnable parameters of the linear projection layer. Taking the CD feature map $\mathbf{P}_{det}$ as the visual input, we build a cross-attention module with e_{text} as the query to realize precise semantic guidance for visual features. The calculation process of the cross-attention is formulated as follows:

$$Attn\left(e_{text}, \mathbf{P}_{det}\right) = \sigma\left(\frac{\left(e_{text}\mathbf{W}^{Q}\right)\left(\mathbf{P}_{det}\mathbf{W}^{K}\right)^{T}}{\sqrt{d}}\right)\mathbf{P}_{det}\mathbf{W}^{V}. \tag{30}$$

The semantically enhanced features output by the cross-attention module are fused with the original CD features via a residual connection, yielding the final optimized feature map

$$\hat{\mathbf{P}}_{det} = LN\left(Attn\left(e_{text}, \mathbf{P}_{det}\right)\cdot\gamma + \mathbf{P}_{det}\right) \tag{31}$$

where γ denotes a learnable scaling factor. $\hat{\mathbf{P}}_{det}$ is then fed into the subsequent detection head to complete pixel-level change prediction. This process leverages the semantic information of core land cover keywords to guide the model to enhance the response weights of changed ground object regions in the CD feature map and effectively improves the localization accuracy of CD.

E. Multitask Training Loss Function

For the CD task, cross-entropy loss is employed to quantify the discrepancy between the predicted change masks and the ground-truth annotations, as formulated in the following:

$$\mathcal{L}_{det} = -\frac{1}{H\times W}\sum_{h=1,w=1}^{H,W}\sum_{j=1}^{C}y_{hw}^{(j)}\log\left(p_{hw}^{(j)}\right) \tag{32}$$

where H and W are the height and width of the change mask and C is the number of classes. At pixel (h, w), the ground-truth label is represented by the one-hot vector $\mathbf{y}_{hw}$, whereas p_{hw} denotes its predicted probability vector.

For CC, the cross-entropy function is likewise employed to measure the discrepancy between each predicted sentence and its corresponding ground-truth sentence

$$\mathcal{L}_{cap} = -\frac{1}{L}\sum_{l=1}^{L}\sum_{v=1}^{V}y_l^{(v)}\log\left(p_l^{(v)}\right) \tag{33}$$

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where L denotes the total count of word tokens and V is the vocabulary size. $y_l^{(v)}$ is a variable flagging whether the ground-truth label belonging to the lth word matches vocabulary index v, whereas $p_l^{(v)}$ gives the predicted probability output by the model for assigning the lth word to such an index.

To reduce the impact of differences in the numerical scales of the two task losses on joint optimization, we adopt a loss normalization strategy based on the detach operation. The overall loss function is formulated as follows:

$$\mathcal{L}_{total} = \frac{\mathcal{L}_{det}}{\text{detach}\left(\mathcal{L}_{det}\right)} + \frac{\mathcal{L}_{cap}}{\text{detach}\left(\mathcal{L}_{cap}\right)} \tag{34}$$

where detach(·) stops gradient propagation through the loss in the denominator. During backpropagation, each denominator is treated as a constant, while the gradient of the corresponding numerator is retained. Consequently, the gradient of each task is scaled by the reciprocal of its current loss magnitude, reducing the influence of scale differences between the CD and CC losses. Compared with learnable loss-weighting methods, this strategy introduces no additional task-weight parameters and requires no weight initialization, providing a simple and parameter-free approach to multitask loss adjustment.

IV. Experiments

A. Data Description

As an extended version of the LEVIR-CC CC benchmark, LEVIR-MCI supplies pixelwise change annotations from CD masks together with semantic descriptive sentences. This dataset records temporal variations captured from 20 different sites located in Texas, USA, over a period of 5–15 years. The dataset comprises 10 077 image pairs accompanied by 50 385 descriptive captions, with every bitemporal image uniformly resized to 256 × 256 pixels. To comprehensively capture the change information between images, each image pair is annotated by five independent annotators to generate five differentiated descriptions, finally forming a large-scale annotation collection covering multidimensional change features.

B. Experimental Setup

1) Evaluation Metrics: For the multiclass remote-sensing image change-detection task, Mean Intersection over Union (MIoU) serves as the core evaluation indicator. This metric achieves objective pixel-level accuracy assessment by calculating the overlapping region between model-generated change masks and ground-truth reference masks. It can be mathematically expressed as

$$\text{IoU}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{35}$$

$$\text{MIoU} = \frac{1}{N}\sum_{i=1}^{N} \text{IoU}_i \tag{36}$$

where N denotes the number of classes. For class i, TP_i, FP_i, and FN_i correspond to the numbers of true positives, false positives, and false negatives, respectively.

The CC task requires metrics to quantify the degree of matching between the captions generated by the model and human reference sentences. To evaluate the proposed model fairly and reliably in terms of its ability to describe changes in bitemporal remote sensing images, four standard image captioning metrics are employed: BLEU-N (N = 1, 2, 3, 4), ROUGE_L, METEOR, and CIDEr, for which larger values correspond to better model performance. In addition, in accordance with [38], we adopt the average metric S_m^* as follows:

$$S_m^* = \frac{1}{4}\left(\text{BLEU-4} + \text{ROUGE_L} + \text{METEOR} + \text{CIDEr}\right). \tag{37}$$

2) Implementation Details: The UniCIC is implemented using the PyTorch deep learning framework. During training, we jointly optimize the CD loss and the caption generation loss. The Adam optimizer is adopted to update model parameters, with an initial learning rate of 1e-4 for both the encoder and the decoder. The maximum number of training epochs is set to 250, the batch size is 16, and the word embedding dimension is 512. After each epoch, model performance is evaluated on the validation set. We reduce the initial learning rate by 50% at every 10-epoch interval. An early-stopping rule is adopted once the combined metric of BLEU-4 and MIoU fails to gain performance gains within 50 consecutive training epochs. We select the best-checkpoint model according to validation-set performance for test-set inference and adopt the average value of five separate runs as our final measurement.

3) Comparison With State-of-the-Art Methods: Currently, there are relatively few research works that can implement the joint processing of the two tasks simultaneously. To fully validate the comprehensive performance of our proposed model on both tasks, we carry out comparative experiments on the LEVIR-MCI dataset: in addition to MCINet and CTMTNet [39], typical multitask baselines, we also select benchmark methods from the respective domains of the two tasks for performance comparison. CD performance is compared with several representative methods: FC-EF [18], FC-SiamConc [18], FC-Siam-Di [18], BIT [40], ACABFNet [41], DARNet [42], DMINet [43], BiFA [44], STRobustNet [45], and ConvFormer-CD [46]. For CC, the proposed model is evaluated against recent advanced approaches, namely, Capt-Rep-Diff [28], Capt-Att [28], Capt-Dual-Att [28], DUDA [28], MCCFormer-S [47], MCCFormer-D [47], RSICCFormer [29], PSNet [48], PromptCC [49], SEN [50], RSCaMa [31], and Change3D [17].

C. Results and Analysis

1) Quantitative Results and Analysis: Comprehensive quantitative comparisons of UniCIC against popular SOTA baselines for single-task and multitask scenarios are summarized in Table I. As clearly demonstrated by the tabulated results, most multitask frameworks achieve remarkably better overall performance than conventional single-task models trained independently, which validates the strong complementarity and effectiveness of collaborative optimization between CD and CC.

Specifically, UniCIC attains the best performance on the CC task. Relative to the single-task CC SOTA model Change3D, the proposed model yields improvements of 1.56%, 0.84%, 1.07%, and 2.38% in BLEU-4, METEOR,

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TABLE I

QUANTITATIVE RESULTS OF THE PROPOSED METHOD AND STATE-OF-THE-ART MODELS ON THE LEVIR-MCI DATASET. TO ENSURE A FAIR COMPARISON, WE FIRST EVALUATE THE PERFORMANCE AGAINST METHODS IN EACH OF THE TWO TASK DOMAINS SEPARATELY AND THEN FURTHER VALIDATE AGAINST THE FUNDAMENTAL JOINT TRAINING METHOD

	Method	MIoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE$_L$	CIDEr-D	S_m^*
Change Detection	FC-EF	0.8270	-	-	-	-	-	-	-	-
	FC-Siam-Conc	0.8425	-	-	-	-	-	-	-	-
	FC-Siam-Di	0.8420	-	-	-	-	-	-	-	-
	BIT	0.8416	-	-	-	-	-	-	-	-
	ACABFNet	0.8443	-	-	-	-	-	-	-	-
	DARNet	0.8436	-	-	-	-	-	-	-	-
	DMINet	0.8482	-	-	-	-	-	-	-	-
	BiFA	0.8525	-	-	-	-	-	-	-	-
	STRobustNet	0.8366	-	-	-	-	-	-	-	-
	ConvFormer-CD	0.8479	-	-	-	-	-	-	-	-
Change Captioning	Capt-Rep-Diff	-	0.7290	0.6198	0.5362	0.4741	0.3447	0.6564	1.1057	0.6452
	Capt-Att	-	0.7764	0.6740	0.5924	0.5315	0.3658	0.6973	1.2122	0.7017
	Capt-Dual-Att	-	0.7951	0.7057	0.6323	0.5746	0.3656	0.7069	1.2442	0.7228
	DUDA	-	0.8144	0.7222	0.6424	0.5779	0.3715	0.7104	1.2432	0.7257
	MCCFormer-S	-	0.7945	0.7164	0.6454	0.5907	0.3653	0.6984	1.2206	0.7187
	MCCFormer-D	-	0.7860	0.6934	0.6171	0.5553	0.3598	0.6871	1.2264	0.7071
	RSICCFormer	-	0.8255	0.7437	0.6751	0.6196	0.3829	0.7178	1.2949	0.7538
	PSNet	-	0.8386	0.7513	0.6789	0.6211	0.3880	0.7360	1.3262	0.7678
	PromptCC	-	0.8277	0.7529	0.6886	0.6323	0.3846	0.7409	1.3526	0.7776
	SEN	-	0.8381	0.7660	0.7039	0.6516	0.3931	0.7424	1.3517	0.7847
	RSCaMa	-	0.8579	0.7799	0.7104	0.6524	0.3991	0.7524	1.3656	0.7924
	Change3D	-	0.8581	0.7781	0.7057	0.6438	0.4003	0.7512	1.3829	0.7946
MCI	CTMTNet	0.8537	0.8595	0.7779	0.7074	0.6469	0.3949	0.7454	1.3494	0.7842
	MCINet	**0.8583**	0.8644	0.7822	0.7113	0.6538	0.4038	0.7578	1.3958	0.8028
	Ours	0.8548	**0.8661**	**0.7888**	**0.7189**	**0.6594**	**0.4087**	**0.7619**	**1.4067**	**0.8091**

ROUGE_L, and CIDEr-D, respectively. Compared to the multitask model MCINet, the improvements on BLEU-4, METEOR, ROUGE_L, and CIDEr-D are 0.56%, 0.49%, 0.41%, and 1.09%, respectively, with the most significant gain observed on CIDEr-D, indicating a higher degree of alignment between the descriptions generated by our proposed model and the human-consensus reference descriptions in terms of key information. On the CD task, the MIoU of our model is slightly lower than that of the current best-performing method. This is attributed to the fact that the spatiotemporal features extracted by our video encoder emphasize temporal semantic modeling, resulting in weaker capability for delineating pixel-level change boundaries compared to dedicated segmentation models. Nevertheless, the detection performance remains at a high level and surpasses other mainstream single-task detection methods. Compared with the substantial gains achieved on the captioning task, this tradeoff aligns with the design objective of our multitask framework, which prioritizes semantic understanding.

2) Qualitative Results and Analysis: Fig. 4 compares the qualitative CD and CC outputs produced by the proposed method and the baseline on the LEVIR-MCI dataset, covering the dual-task outputs of CD and CC, with zoom-in views clearly illustrating fine-grained differences. In the CD task, the baseline exhibits insufficient detection completeness for small-scale change targets and is vulnerable to illumination interference, leading to spurious change responses in nonchange regions. As an illustration, the baseline model misses the newly built constructions in the bottom region of the scene for the first-row sample; in the fourth-row sample, it generates redundant spurious road responses in nonchange regions. By contrast, for newly added land covers, the proposed method substantially enhances detection completeness and produces change-mask contours that closely correspond to the ground-truth annotations. The CC results further highlight the advantages of the proposed method: captions generated by the baseline commonly lack spatial localization information and exhibit semantic inconsistency with the CD results. For instance, in the fourth-row sample, the baseline's mask outputs three change responses corresponding to newly built buildings, while its caption only mentions two building changes, resulting in semantic-spatial misalignment. In contrast, the captions generated by UniCIC are highly aligned with the ground-truth semantics and provide more comprehensive descriptions of scene changes.

The performance improvement in both tasks stems from the synergy between the heterogeneous perceptual frame configuration and the mutual guidance mechanism proposed in this work. Specifically, robust semantic priors are supplied by the SAM-based region-level perceptual frames, enabling

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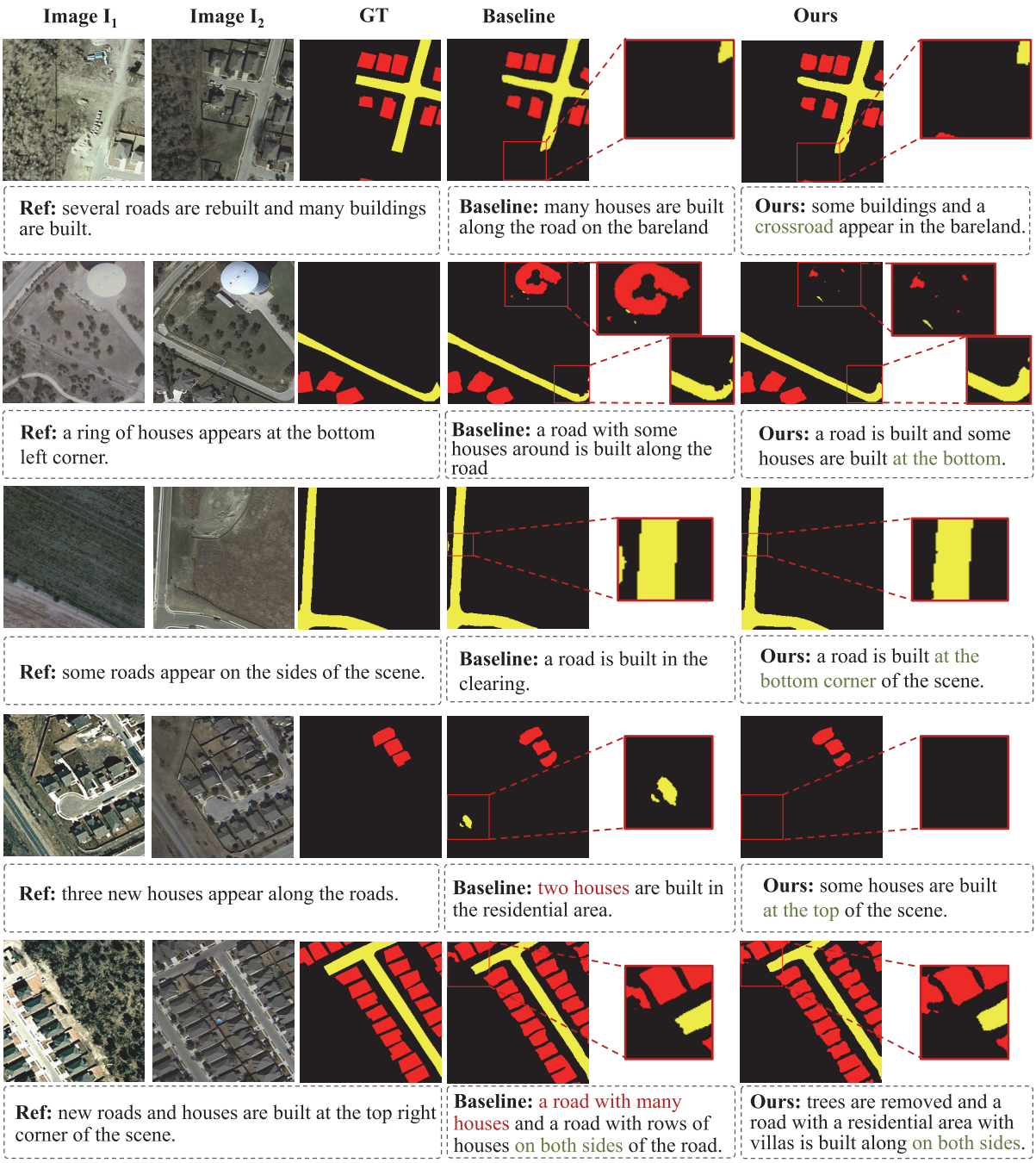


Fig. 4. Qualitative comparison between the baseline and UniCIC on the LEVIR-MCI dataset. The baseline configuration employs only two randomly initialized perception frames and does not include the S^2MG mechanism. In the detection masks, red, yellow, and black denote building changes, road changes, and unchanged background, respectively. In the generated captions, green indicates information consistent with the references, whereas red indicates inaccuracies. The enlarged regions highlight fine-grained differences between the detection results.

the CC branch to attend to meaningful change targets. Meanwhile, the randomly initialized CD frames maintain spatial generalization capability. The S^2MG module further aligns the spatial localization of CD with the semantic understanding of CC, enabling mutual reinforcement between the two tasks. Consequently, pixel-level localization accuracy in CD and semantic reliability in CC are improved simultaneously. The results demonstrate that the proposed method significantly outperforms the baseline in both pixel-level change localization and high-level semantic description, thereby validating the effectiveness of UniCIC for multitask change interpretation.

D. Ablation Study and Analysis

1) Effectiveness of Individual Components of UniCIC: To quantitatively verify the individual effectiveness and collaborative benefits of each core module, a series of ablation variants are designed for comparative experiments. The baseline model only embeds two randomly initialized perceptual frames, without adopting an SAM-based region-level perception frame or

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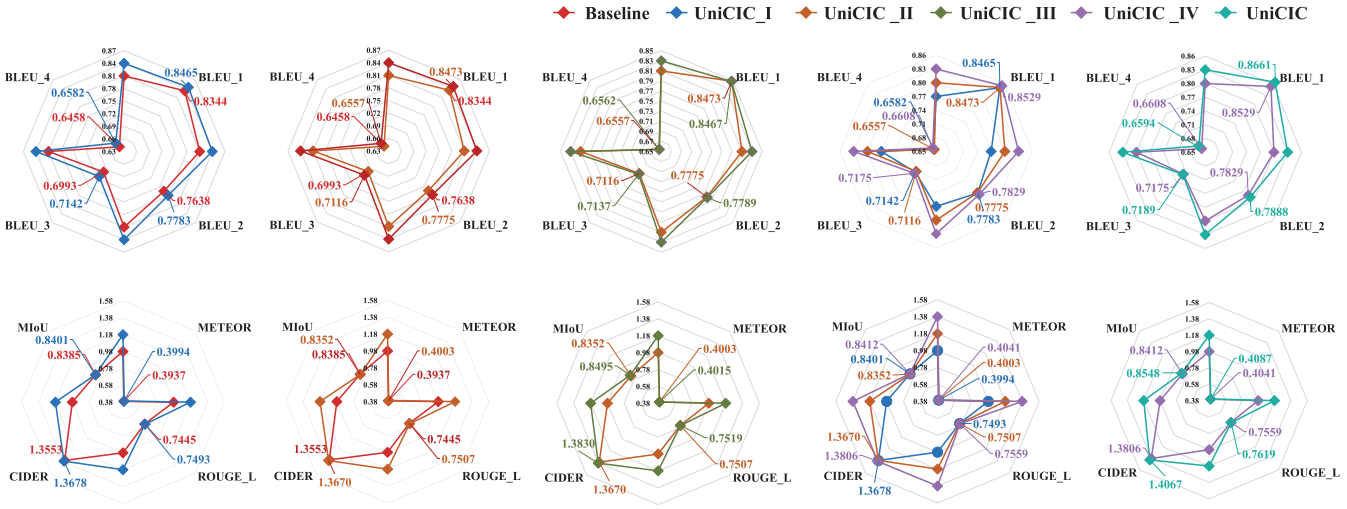


Fig. 5. Effectiveness of individual components of UniCIC.

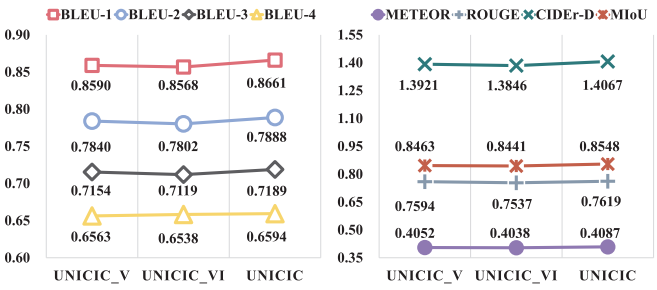


Fig. 6. Quantitative evaluation of the impact of heterogeneous perception frame arrangement on multitask change interpretation performance.

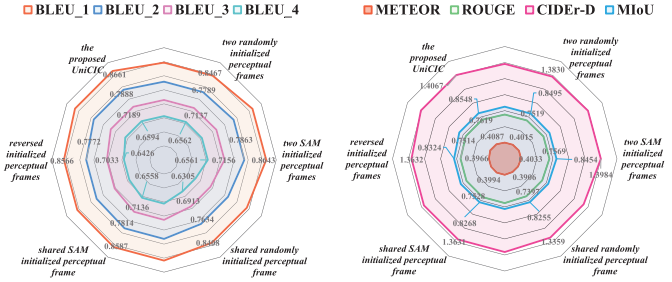


Fig. 7. Quantitative evaluation of the impact of heterogeneous perception frame assignment on multitask change interpretation performance.

the S^{2}MG mechanism. On this basis, key components are added progressively, including the sole introduction of SAM_{RL} (UniCIC_I), one-way guidance from CD to CC (UniCIC_II), exclusive construction of S^{2}MG (UniCIC_III), and the combination of SAM_{RL} with one-way guidance (UniCIC_IV). Such stepwise decomposition of core structures enables clear quantification of the contributions of each module to the overall performance in multitask change interpretation.

As shown in Fig. 5, the introduction of SAM-based region-level perceptual frames yields significant improvements in BLEU-N and CIDEr-D, demonstrating that region-level semantic priors strengthen the feature representation of changed land covers and effectively enhance the semantic accuracy and completeness of the generated captions. Building upon this, the integration of the S^{2}MG mechanism leads to simultaneous optimization of both the localization accuracy of MIoU and CC metrics. This verifies the mutual promotion effect of bidirectional constraints on spatial localization and semantic understanding, which effectively aligns the outputs of the two tasks and reduces semantic noise and pseudochange interference. The complete UniCIC model achieves the best performance on most metrics and consistently outperforms the baseline, demonstrating its overall effectiveness and the complementary benefits of its core components.

2) Effect of Heterogeneous Perception Frame Arrangement: To systematically analyze how different arrangements of heterogeneous perceptual frames affect the overall model performance, we, respectively, design UniCIC_V and UniCIC_VI, which denote arrangement modes $\{\mathbf{I}^P_{\text{Random}}, \mathbf{I}^P_{\text{SAM}}, \mathbf{I}_1, \mathbf{I}_2\}$ and $\{\mathbf{I}_1, \mathbf{I}_2, \mathbf{I}^P_{\text{Random}}, \mathbf{I}^P_{\text{SAM}}\}$, respectively. As shown in Fig. 6, the results prove that embedding perceptual frames in the middle of the temporal sequence obtains the optimal quantitative metrics. This superiority mainly benefits from the VE module, which fully explores spatiotemporal correlation features between the middle perceptual frame and adjacent temporal images. The model can capture interframe differences more efficiently, so as to accurately locate and distinguish changed areas.

3) Impact of Heterogeneous Perception Frame Assignment: To systematically validate the necessity of heterogeneous perception frames and the rationality of their task assignment, we compare the proposed configuration with five perception frame variants, as shown in Fig. 7. First, we evaluate two homogeneous dual-frame configurations. The results show that the dual-random configuration provides limited support for CC semantic representation because it lacks object-level regional priors, whereas the dual-SAM configuration introduces regional priors into CD and may interfere with the learning of fine-grained spatial change information. Second,

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in the single-frame shared configurations, CD and CC share either a randomly initialized perception frame or an SAM-based perception frame. Both shared configurations perform worse than the proposed task-specific dual-frame configuration, indicating that a single perception frame has difficulty simultaneously accommodating the different feature requirements of spatial localization and semantic description. Finally, in the reversed heterogeneous configuration, the SAM-based perception frame is assigned to CD, while the randomly initialized perception frame is assigned to CC. This configuration also performs worse than the proposed assignment, indicating that the performance improvement arises not only from the introduction of heterogeneous information but also from the specific correspondence between the perception frames and the two tasks. Overall, the proposed configuration achieves a better balance between CD and CC performance, validating the necessity and rationality of the task-specific heterogeneous perception frame design.

V. Conclusion

In this article, we propose UniCIC, a unified collaborative inference framework for remote sensing CD and captioning. Its SP-TOPE module jointly encodes the bitemporal images with two heterogeneous perception frames, namely, an SAM-prior-based region-level perception frame for CC and a randomly initialized learnable perception frame for CD, thereby forming task-oriented change representations for semantic interpretation and spatial localization during feature encoding. In addition, the proposed S^2MG mechanism establishes prediction-level cross-task interactions between CD and CC, promoting complementary learning and collaborative inference while improving the consistency between change localization and semantic description. Experiments on the LEVIR-MCI dataset demonstrate that UniCIC achieves state-of-the-art CC performance and competitive CD accuracy, validating the effectiveness of task-oriented heterogeneous encoding and cross-task collaborative interaction. In future work, we plan to explore extending UniCIC from bitemporal image pairs to multitemporal remote sensing sequences. By modeling temporal dependencies across multiple observations, the extended framework may capture the continuous evolution of land-cover changes and provide more comprehensive descriptions of their temporal progression.

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